

1. Setup e Imports

```
# Instalar SAELens si no está disponible:
# !pip install sae-lens

import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import math
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../..'))
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkpoint_dir, get_model_dir
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, get_logs_dir

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4, suppress=True)

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Dispositivo: {device}')
```

Dispositivo: cuda

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'topk',                # Arquitectura del SAE
    'tecnica': 'batch-topk',          # BatchTopK
    'capa': 6,                        # Layer del modelo OthelloGPT
    'juegos': 25000,                  # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',
    'experiment_base_path': os.path.abspath('../..../experiments'),
    'extras': {
        'expansion': 8,               # d_sae / d_in = 8192 / 512
        'k': 50,                      # número de features activas L0 promedio   k
        'epochs': 1000,
        'patience': 20
    }
}

print('\nMetadata de naming:')
for key, value in NAMING_META.items():
    print(f'  {key}: {value}')
```

Metadata de naming:

```
variante: topk
tecnica: batch-topk
capa: 6
juegos: 25000
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\models\experiments
extras: {'expansion': 8, 'k': 50, 'epochs': 1000, 'patience': 20}
```

```
extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
    NAMING_META['extras']
```

```

) if NAMING_META.get('extras') else None

# get_plots_dir crea el directorio automáticamente
plots_dir = Path(get_plots_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
))

print(f'extras_id: {extras_id}')
print(f'plots_dir: {plots_dir}')

```

Nuevo extras registrado: ex3 → {'expansion': 8, 'k': 50, 'epochs': 1000, 'patience': 20}
extras_id: ex3
plots_dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\models\experiments\layer_06\sae_topk_topk\25000games\ex3\plots

3. Configuración del SAE con SAELens

Configuración usando LanguageModelSAERunnerConfig de SAELens con `architecture='batchtopk'`.

Arquitectura BatchTopK: - **BatchTopK activation:** calcula un threshold global por batch para que el **promedio** de features activas por muestra sea **k** (L0 promedio **k**, pero varía por muestra) - **Sin coeficiente de sparsidad:** la sparsity se controla directamente con **k** - **Loss = MSE únicamente** (sin penalización L1)

Parámetro	L1 (standard)	TopK	BatchTopK
Sparsity	sparsity_coef (soft)	k (hard, por muestra)	k (promedio del batch)
L0	Variable	Fijo = k por muestra	Variable por muestra, promedio k
Loss	MSE + $\cdot$ L1	MSE	MSE

Nota sobre el runner: `LanguageModelSAETrainingRunner(cfg).run()` requiere que Othello-GPT esté cargado vía `TransformerLens`. Las celdas siguientes entrenan el SAE directamente sobre los archivos `.npy` pre-extraídos usando la arquitectura `BatchTopK` definida en el config.

```

NUM_EPOCHS = NAMING_META['extras']['epochs']

cfg = {
    'architecture':      'batchtopk',
    'd_in':              512,
    'd_sae':             4096,
    'k':                 50,
    'lr':               1e-3,
    'train_batch_size_tokens': 1024,
    'hook_name':         'blocks.5.hook_resid_post',
}

cfg_summary = pd.DataFrame({
    'Parámetro SAELens': ['architecture', 'd_in', 'd_sae', 'k (L0 promedio)', 'hook_name', 'lr'],
    'Valor': [cfg['architecture'], cfg['d_in'], cfg['d_sae'], cfg['k'],
              cfg['hook_name'], cfg['lr']]
})

print('Configuración SAELens:')
print(cfg_summary.to_string(index=False))

```

Configuración SAELens:

Parámetro SAELens	Valor
architecture	batchtopk
d_in	512
d_sae	4096
k (L0 promedio)	50
hook_name	blocks.5.hook_resid_post
lr	0.001

4. Dataset de Activaciones (Train/Validation Split 80-20%)

Activaciones pre-extraídas con `sae/activations/run_extraction.py` desde la capa `blocks.5.hook_resid_post`.

```

class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""
    def __init__(self, activations_path):
        self.activations = np.load(activations_path)
    def __len__(self):
        return len(self.activations)
    def __getitem__(self, idx):

```

```

        return torch.tensor(self.activations[idx], dtype=torch.float32)

activations_file = Path(f"../../activations/data/layer{NAMING_META['capa']}-{NAMING_META['id']}")
full_dataset = ActivationsDataset(activations_file)

# Split 80-20 reproducible
train_size = int(0.8 * len(full_dataset))
val_size    = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset, [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

train_loader = DataLoader(train_dataset, batch_size=cfg["train_batch_size_tokens"], shuffle=True)
val_loader   = DataLoader(val_dataset,   batch_size=cfg["train_batch_size_tokens"], shuffle=False)

dataset_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validación', 'Total'],
    'Muestras': [f'{len(train_dataset):,}', f'{len(val_dataset):,}', f'{len(full_dataset):,}'],
    'Shape':     [str(full_dataset.activations.shape), '', ''],
})
print(dataset_info.to_string(index=False))

```

```

Conjunto  Muestras      Shape
Train 1,180,000 (1475000, 512)
Validación 295,000
Total 1,475,000

```

5. Modelo: BatchTopK SAE

Implementación de la arquitectura BatchTopK de SAELens.

Diferencia clave vs ReLU + L1 y vs TopK estándar: - El encoder proyecta a `d_sae=8192` dimensiones (pre-activación) - La activación **BatchTopK** calcula un threshold global del batch (k-ésimo valor promediado); cada muestra puede tener más o menos de `k` features activas - L0 es **variable por muestra**, pero el **promedio del batch** `k` (soft constraint a nivel de batch) - No hay `sparsity_coef` ni cosine annealing - El decoder se normaliza a norma unitaria después de cada paso del optimizador

```

class BatchTopKSAE(nn.Module):
    """

```

```

Sparse Autoencoder con activación BatchTopK.
Equivalente a SAE_Lens architecture='batchtopk'.

Encoder      : d_in -> d_sae (lineal + pre-bias centering)
Activación   : BatchTopK (mantiene el promedio de k features activas por batch)
Decoder      : d_sae -> d_in (columnas con norma unitaria)
"""
def __init__(self, d_in: int, d_sae: int, k: int, k_aux: int = 512):
    super().__init__()
    self.d_in = d_in
    self.d_sae = d_sae
    self.k = k
    self.k_aux = k_aux

    self.W_enc = nn.Parameter(torch.empty(d_in, d_sae))
    self.W_dec = nn.Parameter(torch.empty(d_sae, d_in))
    self.b_dec = nn.Parameter(torch.zeros(d_in))

    nn.init.kaiming_uniform_(self.W_enc)
    nn.init.kaiming_uniform_(self.W_dec)
    self.W_dec.data = self.W_dec.data / self.W_dec.data.norm(dim=-1, keepdim=True)
    self.register_buffer('num_batches_not_active', torch.zeros(d_sae))
    self.n_batches_to_dead = 10 # latente muerto si inactivo por 10 batches seguidos

def encode(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
    """Proyecta a espacio latente con activación BatchTopK"""
    x_cent = x - self.b_dec
    pre_acts = F.relu(x_cent @ self.W_enc)
    return self._topk_activation(pre_acts, threshold=threshold)

def _topk_activation(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
    """BatchTopK: selecciona exactamente k * batch_size activaciones usando scatter."""
    if threshold is None:
        acts_topk = torch.topk(x.flatten(), self.k * x.shape[0], dim=-1)
        return (
            torch.zeros_like(x.flatten())
            .scatter(-1, acts_topk.indices, acts_topk.values)
            .reshape(x.shape)
        )
    return torch.where(x > threshold, x, torch.zeros_like(x))

def decode(self, hidden: torch.Tensor) -> torch.Tensor:

```

```

        return hidden @ self.W_dec + self.b_dec

def preprocess_input(self, x: torch.Tensor):
    x_mean = x.mean(dim=-1, keepdim=True)
    x = x - x_mean
    x_std = x.std(dim=-1, keepdim=True)
    x = x / (x_std + 1e-5)
    return x, x_mean, x_std

def postprocess_output(self, x_reconstruct: torch.Tensor,
                       x_mean: torch.Tensor, x_std: torch.Tensor) -> torch.Tensor:
    return x_reconstruct * x_std + x_mean

def forward(self, x: torch.Tensor, threshold=None):
    x_norm, x_mean, x_std = self.preprocess_input(x)
    x_cent = x_norm - self.b_dec
    pre_acts = F.relu(x_cent @ self.W_enc)
    hidden = self._topk_activation(pre_acts, threshold=threshold)
    recon = self.postprocess_output(self.decode(hidden), x_mean, x_std)
    return recon, hidden, pre_acts

def auxiliary_loss(self, x: torch.Tensor, x_reconstruct: torch.Tensor, acts: torch.Tensor):
    """L_aux del repo oficial: reconstruye el residual con top-k_aux dead latents."""
    dead_features = self.num_batches_not_active >= self.n_batches_to_dead
    if dead_features.sum() == 0:
        return torch.tensor(0.0, device=x.device)
    residual = x.float() - x_reconstruct.float()
    acts_topk_aux = torch.topk(
        acts[:, dead_features],
        min(self.k_aux, int(dead_features.sum().item()))),
        dim=-1,
    )
    acts_aux = torch.zeros_like(acts[:, dead_features]).scatter(
        -1, acts_topk_aux.indices, acts_topk_aux.values
    )
    x_reconstruct_aux = acts_aux @ self.W_dec[dead_features]
    return (1/32) * (x_reconstruct_aux.float() - residual.float()).pow(2).mean()

@torch.no_grad()
def update_inactive_features(self, acts):
    self.num_batches_not_active += (acts.sum(0) == 0).float()
    self.num_batches_not_active[acts.sum(0) > 0] = 0

```

```

@torch.no_grad()
def estimate_threshold(self, loader, device, n_batches: int = 10) -> float:
    """Estima un threshold estable promediando sobre múltiples batches
    (independiente del batch actual - para usar en validación/inferencia)."""
    self.eval()
    thresholds = []
    for i, batch in enumerate(loader):
        if i >= n_batches:
            break
        x = batch.to(device)
        x_norm, _, _ = self.preprocess_input(x)
        x_cent = x_norm - self.b_dec
        pre_acts = F.relu(x_cent @ self.W_enc)
        mins = []
        for sample_idx in range(pre_acts.shape[0]):
            pos = pre_acts[sample_idx][pre_acts[sample_idx] > 0]
            if pos.numel() > 0:
                mins.append(pos.min().item())
        t = float(np.mean(mins)) if mins else 0.0
        thresholds.append(t)
    return float(np.mean(thresholds))

@torch.no_grad()
def make_decoder_weights_and_grad_unit_norm(self):
    """Proyecta el gradiente del decoder y renormaliza - igual que el repo oficial."""
    W_dec_normed = self.W_dec / self.W_dec.norm(dim=-1, keepdim=True)
    W_dec_grad_proj = (self.W_dec.grad * W_dec_normed).sum(-1, keepdim=True) * W_dec_normed
    self.W_dec.grad -= W_dec_grad_proj
    self.W_dec.data = W_dec_normed

# Crear modelo desde parámetros del cfg SAELens
model = BatchTopKSAE(d_in=cfg['d_in'], d_sae=cfg['d_sae'], k=cfg['k']).to(device)
num_params = sum(p.numel() for p in model.parameters())

model_info = pd.DataFrame({
    'Componente': [
        'Input', 'Encoder (W_enc)', 'Activación', 'Hidden activos', 'Decoder (W_dec)', 'Output'
    ],
    'k_aux': 'Total params'
},
    'Dimensión': [
        f'{cfg["d_in"]}',

```



```

        f'{cfg["d_in"]} -> {cfg["d_sae"]}',
        f'BatchTopK (k={cfg["k"]})',
        f'{cfg["d_sae"]} (solo {cfg["k"]} activos en promedio)',
        f'{cfg["d_sae"]} -> {cfg["d_in"]}',
        f'{cfg["d_in"]}',
        f'{model.k_aux}',
        f'{num_params:,'}
    ]
})
print(model_info.to_string(index=False))

```

Componente	Dimensión
Input	512
Encoder (W_enc)	512 -> 4096
Activación	BatchTopK (k=50)
Hidden activos	4096 (solo 50 activos en promedio)
Decoder (W_dec)	4096 -> 512
Output	512
k_aux	512
Total params	4,194,816

6. Función de Pérdida BatchTopK

Loss = MSE(reconstrucción)

- **MSE**: error de reconstrucción (igual que en L1)
- **Sin L1 penalty**: la sparsity ya está controlada por el threshold global de BatchTopK
- No hay `sparsity_coef` ni annealing — el único hiperparámetro de sparsity es `k` (promedio del batch)

```

ALPHA_AUX = 1 / 32 # peso de la pérdida auxiliar (arXiv:2412.06410)

def loss_function(x, reconstruction, acts, model):
    """
    Pérdida BatchTopK: MSE + alpha * L_aux (arXiv:2412.06410).
    - MSE      : error de reconstrucción
    - L_aux    : reconstruye el residual con las top-k_aux features para
                  evitar features muertas (alpha = 1/32)

    Args:
        x: Activaciones originales
    """

```

```

        reconstruction: Activaciones reconstruidas
        acts:           Pre-activaciones del encoder
        model:          Instancia de BatchTopKSAE
Returns:
    total_loss (scalar)
"""
mse    = F.mse_loss(reconstruction, x)
l_aux  = model.auxiliary_loss(x, reconstruction, acts)
return mse + l_aux

optimizer = torch.optim.Adam(model.parameters(), lr=cfg["lr"], betas=(0.9, 0.99))

optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning rate', 'betas', 'k (L0 promedio k)', 'k_aux', 'alpha_aux'],
    'Valor':      ['Adam', cfg["lr"], '(0.9, 0.99)', cfg["k"], model.k_aux, f'{ALPHA_AUX:.5f}']
})
print(optim_info.to_string(index=False))

```

	Parámetro	Valor
	Optimizador	Adam
	Learning rate	0.001
	betas	(0.9, 0.99)
k (L0 promedio k)		50
	k_aux	512
	alpha_aux	0.03125 (1/32)

7. Loop de Entrenamiento

```

PATIENCE = NAMING_META['extras']['patience']

save_dir = Path('./saved_model')
save_dir.mkdir(exist_ok=True)
best_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='best',
    extras_id=extras_id
)

```

```

)
BEST_CKPT = save_dir / best_model_name

history = {
    'train_mse_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_mse_loss': [],
    'val_l0_sparsity': [],
    'val_fraction_active': [],
}

best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

print(f'Iniciando entrenamiento: {NUM_EPOCHS} épocas máximo')
print(f'Early stopping: patience={PATIENCE}')
print(f'k = {cfg["k"]} features activas (L0 promedio k por batch)')
print(f'k_aux = {model.k_aux} | alpha_aux = {ALPHA_AUX:.5f} (1/32)')
print(f'Best checkpoint: {BEST_CKPT}')
print('=' * 70)

for epoch in range(NUM_EPOCHS):

    # TRAIN
    model.train()
    t_mse = t_l0 = t_frac = 0.0

    for batch in train_loader:
        x = batch.to(device)
        optimizer.zero_grad()
        recon, hidden, pre_acts = model(x)
        total_loss = loss_function(x, recon, pre_acts, model)
        total_loss.backward()
        model.make_decoder_weights_and_grad_unit_norm()
        optimizer.step()
        model.update_inactive_features(hidden)

    with torch.no_grad():
        t_mse += F.mse_loss(recon, x).item()
        t_l0 += (hidden > 0).float().sum(dim=1).mean().item()

```

```

        t_frac += (hidden > 0).float().mean().item()

n = len(train_loader)
history['train_mse_loss'].append(t_mse / n)
history['train_l0_sparsity'].append(t_l0 / n)
history['train_fraction_active'].append(t_frac / n)

# VALIDATION (sin threshold - igual que training, L0 k)
model.eval()
v_mse = v_l0 = v_frac = 0.0

with torch.no_grad():
    for batch in val_loader:
        x = batch.to(device)
        recon, hidden, _ = model(x)
        v_mse += F.mse_loss(recon, x).item()
        v_l0 += (hidden > 0).float().sum(dim=1).mean().item()
        v_frac += (hidden > 0).float().mean().item()

nv = len(val_loader)
history['val_mse_loss'].append(v_mse / nv)
history['val_l0_sparsity'].append(v_l0 / nv)
history['val_fraction_active'].append(v_frac / nv)

print(f"Época {epoch+1:3d}/{NUM_EPOCHS} | "
      f"Train MSE: {history['train_mse_loss'][-1]:.6f} | "
      f"Val MSE: {history['val_mse_loss'][-1]:.6f} | "
      f"Train L0: {history['train_l0_sparsity'][-1]:.1f} | "
      f"Val L0: {history['val_l0_sparsity'][-1]:.1f}")

# Early stopping
if history['val_mse_loss'][-1] < best_val_mse:
    best_val_mse = history['val_mse_loss'][-1]
    best_epoch = epoch + 1
    epochs_no_improve = 0

torch.save(model.state_dict(), BEST_CKPT)
save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],

```

```

        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best',
        extras=NAMING_META['extras'],
        config=cfg,
        history=history,
        optimizer=optimizer
    )
    print(f'  Mejor modelo guardado (época {best_epoch} | Val MSE: {best_val_mse:.6f})')
else:
    epochs_no_improve += 1
    if epochs_no_improve >= PATIENCE:
        print(f'\nEarly stopping en época {epoch+1} (mejor: época {best_epoch})')
        break

# Restaurar mejor modelo
model.load_state_dict(torch.load(BEST_CKPT, weights_only=True))
print(f'\nMejor época: {best_epoch} | Mejor Val MSE: {best_val_mse:.6f}')

# Estimar threshold de inferencia al final del entrenamiento
inference_threshold = model.estimate_threshold(train_loader, device, n_batches=10)
print(f'Threshold de inferencia estimado: {inference_threshold:.4f}')

```

Iniciando entrenamiento: 1000 épocas máximo

Early stopping: patience=20

k = 50 features activas (L0 promedio k por batch)

k_aux = 512 | alpha_aux = 0.03125 (1/32)

Best checkpoint: saved_model\sae_topk_batch-topk_16_25000g_ex3_best.pt

=====

Época 1/1000 | Train MSE: 0.239780 | Val MSE: 0.189838 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt

Mejor modelo guardado (época 1 | Val MSE: 0.189838)

Época 2/1000 | Train MSE: 0.181367 | Val MSE: 0.177177 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt

Mejor modelo guardado (época 2 | Val MSE: 0.177177)

Época 3/1000 | Train MSE: 0.173343 | Val MSE: 0.172448 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt

Mejor modelo guardado (época 3 | Val MSE: 0.172448)

Época 4/1000 | Train MSE: 0.169506 | Val MSE: 0.170082 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt
Mejor modelo guardado (época 4 | Val MSE: 0.170082)
Época 5/1000 | Train MSE: 0.167402 | Val MSE: 0.168520 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt
Mejor modelo guardado (época 5 | Val MSE: 0.168520)
Época 6/1000 | Train MSE: 0.166002 | Val MSE: 0.167470 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt
Mejor modelo guardado (época 6 | Val MSE: 0.167470)
Época 7/1000 | Train MSE: 0.164997 | Val MSE: 0.166792 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt
Mejor modelo guardado (época 7 | Val MSE: 0.166792)
Época 8/1000 | Train MSE: 0.164212 | Val MSE: 0.166217 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt
Mejor modelo guardado (época 8 | Val MSE: 0.166217)
Época 9/1000 | Train MSE: 0.163622 | Val MSE: 0.165783 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt
Mejor modelo guardado (época 9 | Val MSE: 0.165783)
Época 10/1000 | Train MSE: 0.163130 | Val MSE: 0.165497 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt
Mejor modelo guardado (época 10 | Val MSE: 0.165497)
Época 11/1000 | Train MSE: 0.162736 | Val MSE: 0.165027 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt
Mejor modelo guardado (época 11 | Val MSE: 0.165027)
Época 12/1000 | Train MSE: 0.162422 | Val MSE: 0.164912 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt
Mejor modelo guardado (época 12 | Val MSE: 0.164912)
Época 13/1000 | Train MSE: 0.162158 | Val MSE: 0.164661 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt
Mejor modelo guardado (época 13 | Val MSE: 0.164661)
Época 14/1000 | Train MSE: 0.161936 | Val MSE: 0.164535 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt
Mejor modelo guardado (época 14 | Val MSE: 0.164535)

Época 15/1000 | Train MSE: 0.161722 | Val MSE: 0.164341 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 15 | Val MSE: 0.164341)

Época 16/1000 | Train MSE: 0.161554 | Val MSE: 0.164178 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 16 | Val MSE: 0.164178)

Época 17/1000 | Train MSE: 0.161388 | Val MSE: 0.164029 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 17 | Val MSE: 0.164029)

Época 18/1000 | Train MSE: 0.161244 | Val MSE: 0.163991 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 18 | Val MSE: 0.163991)

Época 19/1000 | Train MSE: 0.161116 | Val MSE: 0.163780 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 19 | Val MSE: 0.163780)

Época 20/1000 | Train MSE: 0.160991 | Val MSE: 0.163792 | Train L0: 50.0 | Val L0: 50.0

Época 21/1000 | Train MSE: 0.160887 | Val MSE: 0.163701 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 21 | Val MSE: 0.163701)

Época 22/1000 | Train MSE: 0.160799 | Val MSE: 0.163693 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 22 | Val MSE: 0.163693)

Época 23/1000 | Train MSE: 0.160711 | Val MSE: 0.163672 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 23 | Val MSE: 0.163672)

Época 24/1000 | Train MSE: 0.160619 | Val MSE: 0.163526 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 24 | Val MSE: 0.163526)

Época 25/1000 | Train MSE: 0.160542 | Val MSE: 0.163463 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 25 | Val MSE: 0.163463)

Época 26/1000 | Train MSE: 0.160481 | Val MSE: 0.163543 | Train L0: 50.0 | Val L0: 50.0

Época 27/1000 | Train MSE: 0.160440 | Val MSE: 0.163479 | Train L0: 50.0 | Val L0: 50.0

Época 28/1000 | Train MSE: 0.160368 | Val MSE: 0.163320 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 28 | Val MSE: 0.163320)
 Época 29/1000 | Train MSE: 0.160314 | Val MSE: 0.163270 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 29 | Val MSE: 0.163270)
 Época 30/1000 | Train MSE: 0.160278 | Val MSE: 0.163313 | Train L0: 50.0 | Val L0: 50.0
 Época 31/1000 | Train MSE: 0.160221 | Val MSE: 0.163314 | Train L0: 50.0 | Val L0: 50.0
 Época 32/1000 | Train MSE: 0.160177 | Val MSE: 0.163216 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 32 | Val MSE: 0.163216)
 Época 33/1000 | Train MSE: 0.160130 | Val MSE: 0.163231 | Train L0: 50.0 | Val L0: 50.0
 Época 34/1000 | Train MSE: 0.160097 | Val MSE: 0.163242 | Train L0: 50.0 | Val L0: 50.0
 Época 35/1000 | Train MSE: 0.160056 | Val MSE: 0.163105 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 35 | Val MSE: 0.163105)
 Época 36/1000 | Train MSE: 0.160029 | Val MSE: 0.163077 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 36 | Val MSE: 0.163077)
 Época 37/1000 | Train MSE: 0.160003 | Val MSE: 0.163183 | Train L0: 50.0 | Val L0: 50.0
 Época 38/1000 | Train MSE: 0.159961 | Val MSE: 0.163112 | Train L0: 50.0 | Val L0: 50.0
 Época 39/1000 | Train MSE: 0.159934 | Val MSE: 0.163067 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 39 | Val MSE: 0.163067)
 Época 40/1000 | Train MSE: 0.159901 | Val MSE: 0.163102 | Train L0: 50.0 | Val L0: 50.0
 Época 41/1000 | Train MSE: 0.159882 | Val MSE: 0.163054 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 41 | Val MSE: 0.163054)
 Época 42/1000 | Train MSE: 0.159856 | Val MSE: 0.162972 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 42 | Val MSE: 0.162972)
 Época 43/1000 | Train MSE: 0.159821 | Val MSE: 0.162968 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 43 | Val MSE: 0.162968)

Época 44/1000 | Train MSE: 0.159806 | Val MSE: 0.163002 | Train L0: 50.0 | Val L0: 50.0
 Época 45/1000 | Train MSE: 0.159782 | Val MSE: 0.163111 | Train L0: 50.0 | Val L0: 50.0
 Época 46/1000 | Train MSE: 0.159756 | Val MSE: 0.162937 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 46 | Val MSE: 0.162937)
 Época 47/1000 | Train MSE: 0.159732 | Val MSE: 0.162965 | Train L0: 50.0 | Val L0: 50.0
 Época 48/1000 | Train MSE: 0.159723 | Val MSE: 0.163023 | Train L0: 50.0 | Val L0: 50.0
 Época 49/1000 | Train MSE: 0.159686 | Val MSE: 0.162972 | Train L0: 50.0 | Val L0: 50.0
 Época 50/1000 | Train MSE: 0.159676 | Val MSE: 0.162974 | Train L0: 50.0 | Val L0: 50.0
 Época 51/1000 | Train MSE: 0.159657 | Val MSE: 0.163004 | Train L0: 50.0 | Val L0: 50.0
 Época 52/1000 | Train MSE: 0.159641 | Val MSE: 0.162930 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 52 | Val MSE: 0.162930)
 Época 53/1000 | Train MSE: 0.159626 | Val MSE: 0.162891 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 53 | Val MSE: 0.162891)
 Época 54/1000 | Train MSE: 0.159610 | Val MSE: 0.162875 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 54 | Val MSE: 0.162875)
 Época 55/1000 | Train MSE: 0.159583 | Val MSE: 0.162917 | Train L0: 50.0 | Val L0: 50.0
 Época 56/1000 | Train MSE: 0.159568 | Val MSE: 0.162824 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 56 | Val MSE: 0.162824)
 Época 57/1000 | Train MSE: 0.159547 | Val MSE: 0.162857 | Train L0: 50.0 | Val L0: 50.0
 Época 58/1000 | Train MSE: 0.159540 | Val MSE: 0.162975 | Train L0: 50.0 | Val L0: 50.0
 Época 59/1000 | Train MSE: 0.159521 | Val MSE: 0.162824 | Train L0: 50.0 | Val L0: 50.0
 Época 60/1000 | Train MSE: 0.159515 | Val MSE: 0.162872 | Train L0: 50.0 | Val L0: 50.0
 Época 61/1000 | Train MSE: 0.159498 | Val MSE: 0.162801 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 61 | Val MSE: 0.162801)
 Época 62/1000 | Train MSE: 0.159487 | Val MSE: 0.162824 | Train L0: 50.0 | Val L0: 50.0
 Época 63/1000 | Train MSE: 0.159479 | Val MSE: 0.162827 | Train L0: 50.0 | Val L0: 50.0
 Época 64/1000 | Train MSE: 0.159465 | Val MSE: 0.162821 | Train L0: 50.0 | Val L0: 50.0
 Época 65/1000 | Train MSE: 0.159451 | Val MSE: 0.162867 | Train L0: 50.0 | Val L0: 50.0
 Época 66/1000 | Train MSE: 0.159440 | Val MSE: 0.162766 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt

Mejor modelo guardado (época 66 | Val MSE: 0.162766)

Época	67/1000	Train MSE: 0.159420	Val MSE: 0.162832	Train L0: 50.0	Val L0: 50.0
Época	68/1000	Train MSE: 0.159418	Val MSE: 0.162780	Train L0: 50.0	Val L0: 50.0
Época	69/1000	Train MSE: 0.159403	Val MSE: 0.162807	Train L0: 50.0	Val L0: 50.0
Época	70/1000	Train MSE: 0.159399	Val MSE: 0.162858	Train L0: 50.0	Val L0: 50.0
Época	71/1000	Train MSE: 0.159389	Val MSE: 0.162775	Train L0: 50.0	Val L0: 50.0
Época	72/1000	Train MSE: 0.159375	Val MSE: 0.162934	Train L0: 50.0	Val L0: 50.0
Época	73/1000	Train MSE: 0.159371	Val MSE: 0.162768	Train L0: 50.0	Val L0: 50.0
Época	74/1000	Train MSE: 0.159358	Val MSE: 0.162809	Train L0: 50.0	Val L0: 50.0
Época	75/1000	Train MSE: 0.159347	Val MSE: 0.162744	Train L0: 50.0	Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt

Mejor modelo guardado (época 75 | Val MSE: 0.162744)

Época	76/1000	Train MSE: 0.159344	Val MSE: 0.162875	Train L0: 50.0	Val L0: 50.0
Época	77/1000	Train MSE: 0.159329	Val MSE: 0.162737	Train L0: 50.0	Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt

Mejor modelo guardado (época 77 | Val MSE: 0.162737)

Época	78/1000	Train MSE: 0.159324	Val MSE: 0.162721	Train L0: 50.0	Val L0: 50.0
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Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt

Mejor modelo guardado (época 78 | Val MSE: 0.162721)

Época	79/1000	Train MSE: 0.159305	Val MSE: 0.162741	Train L0: 50.0	Val L0: 50.0
Época	80/1000	Train MSE: 0.159303	Val MSE: 0.162820	Train L0: 50.0	Val L0: 50.0
Época	81/1000	Train MSE: 0.159295	Val MSE: 0.162698	Train L0: 50.0	Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt

Mejor modelo guardado (época 81 | Val MSE: 0.162698)

Época	82/1000	Train MSE: 0.159297	Val MSE: 0.162864	Train L0: 50.0	Val L0: 50.0
Época	83/1000	Train MSE: 0.159282	Val MSE: 0.162679	Train L0: 50.0	Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt

Mejor modelo guardado (época 83 | Val MSE: 0.162679)

Época	84/1000	Train MSE: 0.159278	Val MSE: 0.162777	Train L0: 50.0	Val L0: 50.0
Época	85/1000	Train MSE: 0.159270	Val MSE: 0.162835	Train L0: 50.0	Val L0: 50.0
Época	86/1000	Train MSE: 0.159254	Val MSE: 0.162760	Train L0: 50.0	Val L0: 50.0
Época	87/1000	Train MSE: 0.159251	Val MSE: 0.162703	Train L0: 50.0	Val L0: 50.0
Época	88/1000	Train MSE: 0.159250	Val MSE: 0.162676	Train L0: 50.0	Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt

Mejor modelo guardado (época 88 | Val MSE: 0.162676)

Época	89/1000	Train MSE: 0.159237	Val MSE: 0.162789	Train L0: 50.0	Val L0: 50.0
Época	90/1000	Train MSE: 0.159221	Val MSE: 0.162692	Train L0: 50.0	Val L0: 50.0

Época 91/1000 | Train MSE: 0.159230 | Val MSE: 0.162880 | Train L0: 50.0 | Val L0: 50.0
 Época 92/1000 | Train MSE: 0.159219 | Val MSE: 0.162683 | Train L0: 50.0 | Val L0: 50.0
 Época 93/1000 | Train MSE: 0.159211 | Val MSE: 0.162742 | Train L0: 50.0 | Val L0: 50.0
 Época 94/1000 | Train MSE: 0.159194 | Val MSE: 0.162669 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 94 | Val MSE: 0.162669)
 Época 95/1000 | Train MSE: 0.159203 | Val MSE: 0.162695 | Train L0: 50.0 | Val L0: 50.0
 Época 96/1000 | Train MSE: 0.159188 | Val MSE: 0.162725 | Train L0: 50.0 | Val L0: 50.0
 Época 97/1000 | Train MSE: 0.159179 | Val MSE: 0.162681 | Train L0: 50.0 | Val L0: 50.0
 Época 98/1000 | Train MSE: 0.159177 | Val MSE: 0.162611 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 98 | Val MSE: 0.162611)
 Época 99/1000 | Train MSE: 0.159160 | Val MSE: 0.162721 | Train L0: 50.0 | Val L0: 50.0
 Época 100/1000 | Train MSE: 0.159157 | Val MSE: 0.162670 | Train L0: 50.0 | Val L0: 50.0
 Época 101/1000 | Train MSE: 0.159153 | Val MSE: 0.162671 | Train L0: 50.0 | Val L0: 50.0
 Época 102/1000 | Train MSE: 0.159149 | Val MSE: 0.162730 | Train L0: 50.0 | Val L0: 50.0
 Época 103/1000 | Train MSE: 0.159146 | Val MSE: 0.162642 | Train L0: 50.0 | Val L0: 50.0
 Época 104/1000 | Train MSE: 0.159142 | Val MSE: 0.162648 | Train L0: 50.0 | Val L0: 50.0
 Época 105/1000 | Train MSE: 0.159136 | Val MSE: 0.162683 | Train L0: 50.0 | Val L0: 50.0
 Época 106/1000 | Train MSE: 0.159139 | Val MSE: 0.162663 | Train L0: 50.0 | Val L0: 50.0
 Época 107/1000 | Train MSE: 0.159118 | Val MSE: 0.162691 | Train L0: 50.0 | Val L0: 50.0
 Época 108/1000 | Train MSE: 0.159123 | Val MSE: 0.162586 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_l6_25000g_ex3_best.pt
 Mejor modelo guardado (época 108 | Val MSE: 0.162586)
 Época 109/1000 | Train MSE: 0.159115 | Val MSE: 0.162687 | Train L0: 50.0 | Val L0: 50.0
 Época 110/1000 | Train MSE: 0.159104 | Val MSE: 0.162646 | Train L0: 50.0 | Val L0: 50.0
 Época 111/1000 | Train MSE: 0.159107 | Val MSE: 0.162760 | Train L0: 50.0 | Val L0: 50.0
 Época 112/1000 | Train MSE: 0.159102 | Val MSE: 0.162645 | Train L0: 50.0 | Val L0: 50.0
 Época 113/1000 | Train MSE: 0.159097 | Val MSE: 0.162670 | Train L0: 50.0 | Val L0: 50.0
 Época 114/1000 | Train MSE: 0.159092 | Val MSE: 0.162650 | Train L0: 50.0 | Val L0: 50.0
 Época 115/1000 | Train MSE: 0.159075 | Val MSE: 0.162632 | Train L0: 50.0 | Val L0: 50.0
 Época 116/1000 | Train MSE: 0.159084 | Val MSE: 0.162702 | Train L0: 50.0 | Val L0: 50.0
 Época 117/1000 | Train MSE: 0.159079 | Val MSE: 0.162691 | Train L0: 50.0 | Val L0: 50.0
 Época 118/1000 | Train MSE: 0.159074 | Val MSE: 0.162651 | Train L0: 50.0 | Val L0: 50.0
 Época 119/1000 | Train MSE: 0.159067 | Val MSE: 0.162620 | Train L0: 50.0 | Val L0: 50.0
 Época 120/1000 | Train MSE: 0.159069 | Val MSE: 0.162677 | Train L0: 50.0 | Val L0: 50.0
 Época 121/1000 | Train MSE: 0.159060 | Val MSE: 0.162666 | Train L0: 50.0 | Val L0: 50.0
 Época 122/1000 | Train MSE: 0.159054 | Val MSE: 0.162668 | Train L0: 50.0 | Val L0: 50.0
 Época 123/1000 | Train MSE: 0.159052 | Val MSE: 0.162695 | Train L0: 50.0 | Val L0: 50.0
 Época 124/1000 | Train MSE: 0.159048 | Val MSE: 0.162582 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt

Mejor modelo guardado (época 124 | Val MSE: 0.162582)

Época 125/1000 | Train MSE: 0.159049 | Val MSE: 0.162651 | Train L0: 50.0 | Val L0: 50.0

Época 126/1000 | Train MSE: 0.159035 | Val MSE: 0.162677 | Train L0: 50.0 | Val L0: 50.0

Época 127/1000 | Train MSE: 0.159038 | Val MSE: 0.162625 | Train L0: 50.0 | Val L0: 50.0

Época 128/1000 | Train MSE: 0.159029 | Val MSE: 0.162570 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt

Mejor modelo guardado (época 128 | Val MSE: 0.162570)

Época 129/1000 | Train MSE: 0.159032 | Val MSE: 0.162664 | Train L0: 50.0 | Val L0: 50.0

Época 130/1000 | Train MSE: 0.159024 | Val MSE: 0.162677 | Train L0: 50.0 | Val L0: 50.0

Época 131/1000 | Train MSE: 0.159020 | Val MSE: 0.162639 | Train L0: 50.0 | Val L0: 50.0

Época 132/1000 | Train MSE: 0.159018 | Val MSE: 0.162659 | Train L0: 50.0 | Val L0: 50.0

Época 133/1000 | Train MSE: 0.159017 | Val MSE: 0.162599 | Train L0: 50.0 | Val L0: 50.0

Época 134/1000 | Train MSE: 0.159010 | Val MSE: 0.162604 | Train L0: 50.0 | Val L0: 50.0

Época 135/1000 | Train MSE: 0.159012 | Val MSE: 0.162691 | Train L0: 50.0 | Val L0: 50.0

Época 136/1000 | Train MSE: 0.158994 | Val MSE: 0.162608 | Train L0: 50.0 | Val L0: 50.0

Época 137/1000 | Train MSE: 0.158996 | Val MSE: 0.162621 | Train L0: 50.0 | Val L0: 50.0

Época 138/1000 | Train MSE: 0.159001 | Val MSE: 0.162596 | Train L0: 50.0 | Val L0: 50.0

Época 139/1000 | Train MSE: 0.158984 | Val MSE: 0.162626 | Train L0: 50.0 | Val L0: 50.0

Época 140/1000 | Train MSE: 0.158985 | Val MSE: 0.162626 | Train L0: 50.0 | Val L0: 50.0

Época 141/1000 | Train MSE: 0.158988 | Val MSE: 0.162555 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt

Mejor modelo guardado (época 141 | Val MSE: 0.162555)

Época 142/1000 | Train MSE: 0.158986 | Val MSE: 0.162684 | Train L0: 50.0 | Val L0: 50.0

Época 143/1000 | Train MSE: 0.158971 | Val MSE: 0.162680 | Train L0: 50.0 | Val L0: 50.0

Época 144/1000 | Train MSE: 0.158972 | Val MSE: 0.162658 | Train L0: 50.0 | Val L0: 50.0

Época 145/1000 | Train MSE: 0.158968 | Val MSE: 0.162628 | Train L0: 50.0 | Val L0: 50.0

Época 146/1000 | Train MSE: 0.158966 | Val MSE: 0.162574 | Train L0: 50.0 | Val L0: 50.0

Época 147/1000 | Train MSE: 0.158960 | Val MSE: 0.162522 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\25000games\sae_topk_batch-topk_16_25000g_ex3_best.pt

Mejor modelo guardado (época 147 | Val MSE: 0.162522)

Época 148/1000 | Train MSE: 0.158956 | Val MSE: 0.162646 | Train L0: 50.0 | Val L0: 50.0

Época 149/1000 | Train MSE: 0.158957 | Val MSE: 0.162585 | Train L0: 50.0 | Val L0: 50.0

Época 150/1000 | Train MSE: 0.158956 | Val MSE: 0.162583 | Train L0: 50.0 | Val L0: 50.0

Época 151/1000 | Train MSE: 0.158951 | Val MSE: 0.162563 | Train L0: 50.0 | Val L0: 50.0

Época 152/1000 | Train MSE: 0.158953 | Val MSE: 0.162670 | Train L0: 50.0 | Val L0: 50.0

Época 153/1000 | Train MSE: 0.158944 | Val MSE: 0.162552 | Train L0: 50.0 | Val L0: 50.0

Época 154/1000 | Train MSE: 0.158940 | Val MSE: 0.162574 | Train L0: 50.0 | Val L0: 50.0

Época 155/1000 | Train MSE: 0.158932 | Val MSE: 0.162569 | Train L0: 50.0 | Val L0: 50.0

Época 156/1000		Train MSE: 0.158936		Val MSE: 0.162636		Train L0: 50.0		Val L0: 50.0
Época 157/1000		Train MSE: 0.158930		Val MSE: 0.162581		Train L0: 50.0		Val L0: 50.0
Época 158/1000		Train MSE: 0.158923		Val MSE: 0.162593		Train L0: 50.0		Val L0: 50.0
Época 159/1000		Train MSE: 0.158923		Val MSE: 0.162562		Train L0: 50.0		Val L0: 50.0
Época 160/1000		Train MSE: 0.158930		Val MSE: 0.162598		Train L0: 50.0		Val L0: 50.0
Época 161/1000		Train MSE: 0.158919		Val MSE: 0.162587		Train L0: 50.0		Val L0: 50.0
Época 162/1000		Train MSE: 0.158917		Val MSE: 0.162646		Train L0: 50.0		Val L0: 50.0
Época 163/1000		Train MSE: 0.158919		Val MSE: 0.162619		Train L0: 50.0		Val L0: 50.0
Época 164/1000		Train MSE: 0.158913		Val MSE: 0.162648		Train L0: 50.0		Val L0: 50.0
Época 165/1000		Train MSE: 0.158904		Val MSE: 0.162562		Train L0: 50.0		Val L0: 50.0
Época 166/1000		Train MSE: 0.158905		Val MSE: 0.162616		Train L0: 50.0		Val L0: 50.0
Época 167/1000		Train MSE: 0.158903		Val MSE: 0.162574		Train L0: 50.0		Val L0: 50.0

Early stopping en época 167 (mejor: época 147)

Mejor época: 147 | Mejor Val MSE: 0.162522

Threshold de inferencia estimado: 0.0025

8. Guardar Modelo

```
# Guardar checkpoint final
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': {
        'architecture': cfg["architecture"],
        'd_in': cfg["d_in"],
        'd_sae': cfg["d_sae"],
        'k': cfg["k"],
        'lr': cfg["lr"],
        'hook_name': cfg["hook_name"],
    },
    'history': history,
}

final_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
```

```

        extras_id=extras_id
    )
    checkpoint_path = save_dir / final_model_name
    torch.save(checkpoint, checkpoint_path)
    print(f'Modelo guardado: {checkpoint_path}')

    save_checkpoint(
        model,
        base_path=NAMING_META['base_path'],
        variante=NAMING_META['variante'],
        tecnica=NAMING_META['tecnica'],
        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='final',
        extras=NAMING_META['extras'],
        config=cfg,
        history=history,
        optimizer=optimizer
    )

```

Modelo guardado: saved_model\sae_topk_batch-topk_l6_25000g_ex3_final.pt

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_batch-topk_l6_25000games\sae_topk_batch-topk_l6_25000g_ex3_final.pt

WindowsPath('C:/Users/Esposa/Documents/Repos/sae-othello-gpt/layer_06/models/sae_topk/sae_topk_batch-topk_l6_25000games/sae_topk_batch-topk_l6_25000g_ex3_final.pt')

9. Métricas básicas del modelo

```

import json

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

```

```

training_metrics = {
    # Calidad de reconstrucción
    'val_mse_final':      history['val_mse_loss'][-1],
    'val_mse_best':      best_val_mse,
    'val_l0_final':      history['val_l0_sparsity'][-1],
    'val_fraction_active_final': history['val_fraction_active'][-1],
    # Contexto del entrenamiento
    'best_epoch':        best_epoch,
    'total_epochs':      len(history['train_mse_loss']),
    'train_mse_final':   history['train_mse_loss'][-1],
    'train_l0_final':    history['train_l0_sparsity'][-1],
    # Hiperparámetros TopK
    'architecture':      cfg["architecture"],
    'k':                 cfg["k"],
    'd_sae':             cfg["d_sae"],
    'd_in':              cfg["d_in"],
    'expansion_factor':  cfg["d_sae"] // cfg["d_in"],
    'hook_name':         cfg["hook_name"],
}

os.makedirs(metrics_dir, exist_ok=True)
metrics_path = Path(metrics_dir) / 'training_metrics.json'
with open(metrics_path, 'w') as f:
    json.dump(training_metrics, f, indent=2)

print(f'Métricas guardadas: {metrics_path}')
print(pd.DataFrame(
    list(training_metrics.items()), columns=['Métrica', 'Valor']
).to_string(index=False))

```

Métricas guardadas: c:\Users\Esposa\Documents\Repos\othello_world\sae\models\experiments\lay
topk_topk\25000games\ex3\metrics\training_metrics.json

Métrica	Valor
val_mse_final	0.162574
val_mse_best	0.162522
val_l0_final	50.0
val_fraction_active_final	0.012207
best_epoch	147
total_epochs	167
train_mse_final	0.158903
train_l0_final	50.0
architecture	batchtopk

k	50
d_sae	4096
d_in	512
expansion_factor	8
hook_name	blocks.5.hook_resid_post

10. Visualización de Resultados

```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# MSE Loss
axes[0].plot(history['train_mse_loss'], label='Train', color='steelblue', alpha=0.8)
axes[0].plot(history['val_mse_loss'], label='Validación', color='tomato', alpha=0.8)
if best_epoch > 0:
    axes[0].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7,
                    label=f'Best (época {best_epoch})')
axes[0].set_title('MSE Loss (Reconstrucción)')
axes[0].set_xlabel('Época')
axes[0].set_ylabel('MSE')
axes[0].legend()
axes[0].grid(True, alpha=0.3)

# L0 sparsity (para BatchTopK el promedio del batch k, pero varía por muestra)
axes[1].plot(history['train_l0_sparsity'], label='Train', color='steelblue', alpha=0.8)
axes[1].plot(history['val_l0_sparsity'], label='Validación', color='tomato', alpha=0.8)
axes[1].axhline(cfg["k"], color='orange', linestyle='--', alpha=0.8, label=f'k={cfg["k"]} (p
axes[1].set_title(f'L0 Sparsity (promedio del batch k={cfg["k"]})')
axes[1].set_xlabel('Época')
axes[1].set_ylabel('Features activas (promedio)')
axes[1].legend()
axes[1].grid(True, alpha=0.3)

# Fracción activa
axes[2].plot(history['train_fraction_active'], label='Train', color='steelblue', alpha=0.8)
axes[2].plot(history['val_fraction_active'], label='Validación', color='tomato', alpha=0.8)
expected_frac = cfg["k"] / cfg["d_sae"]
axes[2].axhline(expected_frac, color='orange', linestyle='--', alpha=0.8,
                label=f'k/d_sae={expected_frac:.4f}')
axes[2].set_title('Fracción Features Activas')
axes[2].set_xlabel('Época')
axes[2].set_ylabel('Fracción activa')
```



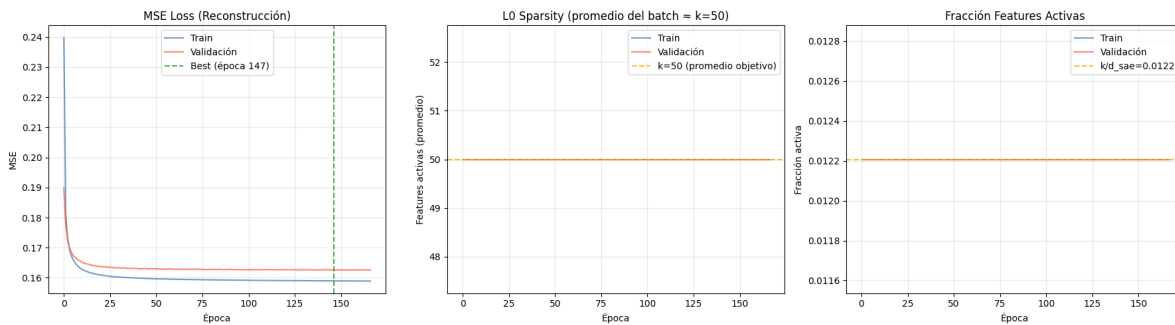
```

axes[2].legend()
axes[2].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(plots_dir / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('Gráficas guardadas exitosamente')
print('='*50)

```



```

=====
Gráficas guardadas exitosamente
=====

```

11. Resumen Final

```

print('\n' + '='*70)
print(' RESUMEN FINAL DEL ENTRENAMIENTO (BatchTopK SAE)')
print('='*70)

arch_summary = pd.DataFrame({
    'Componente': ['Input (d_in)', 'Hidden (d_sae)', 'k (L0 promedio k)', 'Expansión', 'Par
    'Valor': [
        f'{cfg["d_in"]}',
        f'{cfg["d_sae"]}',
        f'{cfg["k"]} features activas (promedio del batch)',
        f'{cfg["d_sae"] // cfg["d_in"]}x',
        f'{num_params:,}'
    ]
})

```

```

    ]
})
print('\nArquitectura:')
print(arch_summary.to_string(index=False))

data_summary = pd.DataFrame({
    'Conjunto': ['Train', 'Validación', 'Total'],
    'Muestras': [f'{len(train_dataset):,}', f'{len(val_dataset):,}', f'{len(full_dataset):,}'],
})
print('\nDatos:')
print(data_summary.to_string(index=False))

final_summary = pd.DataFrame({
    'Métrica': [
        'Val MSE best', 'Val MSE final',
        'Val L0 final (promedio k)', 'Val Fracción activa', 'Mejor época'
    ],
    'Valor': [
        f'{best_val_mse:.6f}',
        f'{history['val_mse_loss'][-1]:.6f}',
        f'{history['val_l0_sparsity'][-1]:.1f} (k={cfg['k']})',
        f'{history['val_fraction_active'][-1]:.4%}',
        f'{best_epoch}/{len(history['train_mse_loss'])}'
    ]
})
print('\nMétricas:')
print(final_summary.to_string(index=False))

```

```

=====
RESUMEN FINAL DEL ENTRENAMIENTO (BatchTopK SAE)
=====

```

Arquitectura:

Componente	Valor
Input (d_in)	512
Hidden (d_sae)	4096
k (L0 promedio k) 50 features activas (promedio del batch)	
Expansión	8x
Parámetros	4,194,816

Datos:

Conjunto	Muestras
Train	1,180,000
Validación	295,000
Total	1,475,000

Métricas:

	Métrica	Valor
	Val MSE best	0.162522
	Val MSE final	0.162574
Val L0 final (promedio k)	50.0 (k=50)	
	Val Fracción activa	1.2207%
	Mejor época	147/167

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```
import subprocess

exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_batch-topk.ipynb'
output_path = exp_dir / pdf_name

print(f'Generando PDF con Quarto...')
```

```

print(f'Archivo de salida: {output_path}')

result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f'PDF generado exitosamente: {output_path}')
else:
    print(f'Error al generar PDF:')
    print(result.stderr)

```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\models\experiments\layer_topk_topk\25000games\ex3\sae_topk_batch-topk_l6_25000g_ex3_training.pdf
 PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\models\experiments\layer_topk_topk\25000games\ex3\sae_topk_batch-topk_l6_25000g_ex3_training.pdf